

PH 528: Introduction to General Relativity

Course Code	PH 528
Course Name	Introduction to General Relativity
L-T-P-C	3-0-0-3
Prerequisites	Mathematical Physics (PH511), Classical Physics (PH 512), Electromagnetic Theory (PH 521).
Intended for	I-Ph.D., M.Sc., B.Tech 3rd and 4th Year.
Mutual Exclusion	None
Approval	46th BoA (May 2022)

Course Contents

- **Special Relativity:** Principles of special relativity – Lorentz transformations, Covariant and contravariant vectors, Relativistic Mechanics. (4 Hours)
- **Tensor Algebra and Tensor Calculus:** Manifolds and metric, Introduction to tensors – Transformation of coordinates, Lie derivatives – covariant differentiation – Christoffel symbols, The Riemann and Ricci tensors – The Bianchi identities, Geodesics, Isometries – The Killing equation and conserved quantities. (9 Hours)
- **General Relativity:** The equivalence principle – The principle of general covariance, The stress-energy tensor, Einstein equations, The equation of geodesic deviation, linearized gravity and idea of gravitational waves. (9 Hours)
- **Schwarzschild solution and Black holes:** The Schwarzschild solution, Motion of particles in the Schwarzschild metric – Precession of the perihelion – Bending of light, Black holes – event horizon and singularity, The Kruskal extension – Penrose diagrams. (10 hours)
- **Cosmology:** Homogeneity and isotropy – The FRW metric, Friedmann equations – Solutions with different types of matter, Cosmological redshift – standard candles, Dark matter and dark energy, Thermal history of the universe, Horizon problem and Inflation. (10 hours)

Text Books

1. J. B. Hartle, **Gravity: An Introduction to Einstein's General Relativity**, Pearson Education India, 2003
2. B. F. Schutz, **A First Course in General Relativity**, 2nd Edition, Cambridge University Press, 2009

References

1. S. Carroll, **Spacetime and Geometry**, Addison Wesley, 2004.
2. Barbara Ryden, **Introduction to Cosmology**, 2nd Edition, Addison-Wesley, 2016.